

Geometry, Entropy, and Operator Learning with Tunable Kernels

Patrick S. Mahon

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Abstract

We study a two-dial family of kernels

$$k_{\theta,\sigma}(y \mid x) \propto w_{\theta}(\|x - y\|) g_{\sigma}(y - x),$$

whose product form covers any radial weight w_{θ} and centred residual density g_{σ} that satisfy minimal smoothness and moment conditions. To give the theory a concrete footing—and to obtain closed-form bias constants in our finite-sample analysis—we instantiate this family with a data-adaptive *local-linear* construction, but all results hold for the broader class.

For every observable f we compute the one-step cumulant-generating function (CGF) and show that its quadratic coefficient $\mathcal{I}_{\theta,\sigma}[f]$ serves simultaneously as (i) a *Fisher load*, (ii) $\frac{2}{\theta^2}$ times the synthetic Ricci curvature $\Gamma_2^{(\theta,\sigma)}(f)$, and (iii) $-2 \partial_{\sigma}$ of the one-step Shannon entropy. This *Fisher-curvature-entropy bridge* yields a single quadratic objective whose minimiser in the dial plane, (θ^*, σ^*) —the *little-rose*—optimises curvature bias, entropy production, and operator-learning risk at once. We prove

$$\nabla_{(\theta,\sigma)} \text{KL}(\mu_1 \parallel k_{\theta,\sigma} \# \mu_0) = -\frac{1}{2} \nabla_{(\theta,\sigma)} \mathcal{I}_{\theta,\sigma},$$

so steepest KL descent converges to the little-rose with no hyper-parameters. A data-driven Goldenshluger–Lepski selector recovers $(\hat{\theta}_{\star}, \hat{\sigma}_{\star})$ from finite samples, and we give non-asymptotic risk bounds for both Perron–Frobenius and Koopman estimates.

The resulting dial matrices $(\Theta_{ij}, \Sigma_{ij})$ form a directed, doubly-weighted graph, opening avenues toward discrete curvature flows, entropy production on cycles, and multiparameter persistent homology—highlighting a promising intersection of statistical dynamics, information geometry, and topological data analysis.

1 Introduction

1.1 Why tune local kernels?

1.2 Key contributions

- Closed-form bridge: CGF curvature $\rightarrow$ Fisher load $\rightarrow$ curvature & entropy.
- Single KL optimisation (little-rose) jointly tunes **two** dials (θ, σ) .
- Finite-sample Goldenshluger–Lepski selector with risk guarantees.

1.3 Organisation of the paper

2 Rose kernels and dials

2.1 Product-form kernels and minimal assumptions (K0–K3)

$$z := y - x, \quad r := \|z\|.$$

(W1) **(Radial weight)** $w : [0, \infty) \rightarrow [0, \infty)$ is C^1 , radial, $\int_{\mathbb{R}^d} w(\|z\|) dz = 1$, $\int \|z\|^2 w(\|z\|) dz < \infty$, and $w \in L^2(\mathbb{R}^d)$.

(W2) **(Centred residual)** $g : \mathbb{R}^d \rightarrow [0, \infty)$ is C^1 , $\int g(z) dz = 1$, $\int z g(z) dz = 0$, $\int z z^\top g(z) dz < \infty$, and $\nabla g \in L^2(\mathbb{R}^d)$.

Definition 2.1 (Rose kernel). For $(\theta, \sigma) \in (0, \infty)^2$ set

$$w_\theta(r) := \theta^{-d} w(r/\theta), \quad g_\sigma(z) := \sigma^{-d} g(z/\sigma), \quad k_{\theta,\sigma}(y | x) := \frac{w_\theta(\|x - y\|) g_\sigma(y - x)}{\int_X w_\theta(\|x - y'\|) g_\sigma(y' - x) dy'}$$

(K1) **(Markov)** $k_{\theta,\sigma}(y | x) \geq 0$ and $\int_X k_{\theta,\sigma}(y | x) dy = 1$.

(K2) **(Score in L^2)** $S_{\theta,\sigma}(y | x) := \nabla_y \log k_{\theta,\sigma}(y | x) \in L^2(dy dx)$.

(K3) **(Second-moment scaling)** $\int_X \|y - x\|^2 k_{\theta,\sigma}(y | x) dy = O(\theta^2)$ as $\theta \downarrow 0$.

(K4) **(C^1 in dials)** $\partial_\theta \log k_{\theta,\sigma}, \partial_\sigma \log k_{\theta,\sigma} \in L^2(dy dx)$.

2.2 Local-linear residual construction (canonical example)

2.3 Scaled kernels and notation for the rest of the paper

3 Population theory

(K0)–(K2) as in Sec. 3.0 — Markov, score L^2 , second-moment scaling.

3.1 CGF $\Rightarrow$ Fisher $\Rightarrow$ Landau

Definition 3.1 (One-step CGF).

Proposition 3.1 (CGF–Fisher correspondence).

Remark 3.1 (Legendre dual = Landau stiffness). A two-line box reminding readers that the Legendre transform of Λ yields a quadratic Landau free energy.

3.2 Fisher–curvature–entropy bridge

Lemma 3.1 (Curvature identity).

Lemma 3.2 (Discrete de Bruijn).

Theorem 3.1 (Fisher–curvature–entropy bridge).

3.3 Introducing the dial plane

(K0) C^1 regularity in dials. The maps $(\theta, \sigma) \mapsto \log k_{\theta, \sigma}$ are $C^1 \dots$

3.4 KL gradient and the *little-rose* optimiser

Theorem 3.2 (KL gradient).

Definition 3.2 (Little-rose).

Corollary 3.1 (Unified optimality).

4 Finite-sample dial selection

4.1 Goldenshluger–Lepski selector

4.2 Risk bounds for PF and Koopman estimates

Theorem 4.1 (GL selector, risk bound).

4.3 Illustrative experiment

5 Discussion and Outlook

5.1 Key implications

(I1) Unified diagnostic of fluctuation strength

The Fisher load $\mathcal{I}_{\theta, \sigma}$ is *simultaneously* the variance coefficient, the synthetic–curvature prefactor, and the entropy–production rate. A single dial pair (θ, σ) governs them all.

(I2) Automatic bias–variance balance

Steepest descent of $\text{KL}(\mu_1 \parallel k_{\theta, \sigma} \# \mu_0)$ is steepest *ascent* of $\mathcal{I}_{\theta, \sigma}$, so the statistically most informative kernel is found with no cross-validation.

(I3) One-step Schrödinger bridge

The little-rose kernel realises the discrete entropic transport $\mu_0 \rightarrow \mu_1$, linking operator learning to optimal-transport theory.

(I4) Landau lens on data

The Legendre transform of the CGF equips any observable with a Landau-type free-energy landscape whose stiffness is data-computable.

5.2 Future directions

(F1) Directed discrete curvature

Dial matrices $(\Theta_{ij}, \Sigma_{ij})$ define a doubly-weighted digraph, opening analytic work on Ollivier/Bakry–Émery curvature for non-reversible kernels.

(F2) Multiparameter persistent homology

The two weights furnish a natural bifiltration, inviting multi-parameter TDA of dynamical data sets.

5.3 Limitations

5.4 Conclusion

A Proof details for Sec. 3

B GL selector derivations

C Additional experiments

D Dial-graph topology (exploratory)